

Daffodil International University

Department of Computer Science & Engineering

AI Innovation Hackathon 2026

Organized By Department Of CSE, DIU

Final Round: Additional Challenge Task

1. TEAM INFORMATION

Team Name: Runtime Terrors

Project Title: ClusterMind, AI-Powered Self-Healing Cluster Intelligence

Track: AI for Cluster Intelligence (Predictive Operations)

2. PROBLEM STATEMENT

ClusterMind can move a workload from a risky node to a healthy one by itself. It promises zero data loss, but that promise is never checked after the move. A risky node can also receive new work while it is still weak. The team must prove that recovery is correct and keep sick nodes out of service.

3. BACKGROUND

Automatic healing is only safe when it can be verified. A restored job that lost steps is worse than a clear crash, because nobody notices. Sending fresh work to a weak node can start the same failure again. Proof of recovery is what allows an operator to step back.

4. ADDITIONAL CHALLENGE TASK

Build a new module named “Recovery Verification and Safe Mode”. It must sit on top of your existing checkpoint and migrate loop.

Recovery Check: After a move, confirm that the job restarts from the correct checkpoint. Report any lost steps or lost seconds.

Safe Mode: Block new work from a risky node until its health returns to normal. Show the node state on the dashboard.

Success Report: Show the migration success rate, the average recovery time and the number of false alarms.

Add one simple sentence to each audit entry that explains the action. Use simulated telemetry if the real cluster is not available.

5. EXPECTED DELIVERABLES

After two and a half hours, teams must show an updated prototype in a live or simulated run. The run must show one verified recovery and one node placed in safe mode. The dashboard must show the success report. Teams must submit the source code, a short note on their design choices, and a few slides.

6. EVALUATION CONSIDERATIONS

Judges may assess innovation, technical quality, and the quality of integration with the existing system. They may also assess real user impact, honest and effective use of AI tools, and the clarity of the final presentation.